
Student Performance Prediction System

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Student Performance Prediction System

Phase-3 Project Report

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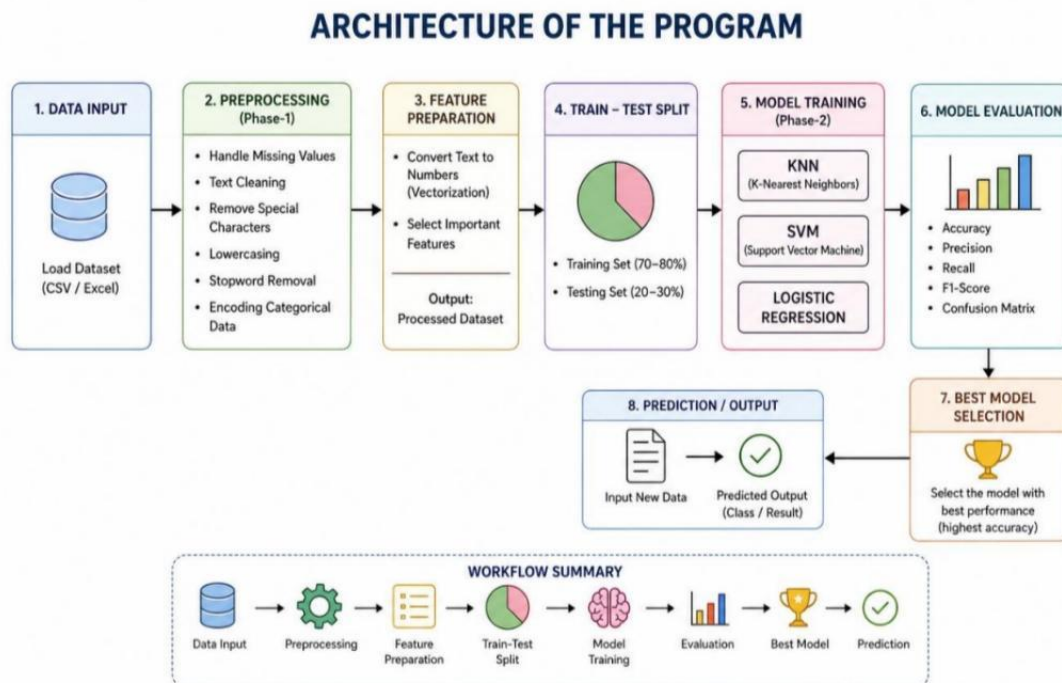
1. Problem Description

The Student Performance Prediction System is developed to analyze and predict student academic performance using machine learning techniques. The system uses student-related features such as study hours, attendance, internal marks, previous results, sleep hours, and extracurricular activities to predict final scores and pass/fail status.

1.1 Dataset Description

- study_hours – Number of hours spent studying.
- attendance – Attendance percentage of students.
- internal_score – Internal assessment marks.
- prev_result – Previous academic result.
- sleep_hours – Average sleeping hours.
- extra_activity – Participation in extracurricular activities.
- final_score – Final obtained marks.
- pass_fail – Pass or fail category.

2. Flow diagram



3. Modules Used

Modules for Project Implementation

2.1 Modules Used

The following inbuilt and external Python modules were used in this project:

1. NumPy

- Used for numerical operations and handling arrays.
- Helps in combining feature vectors and ratings efficiently.

2. Pandas

- Used for data manipulation and analysis.
- Helps in reading datasets, cleaning data, and handling dataframes.

3. Scikit-learn (sklearn)

- A major library used for machine learning tasks:

3. Machine Learning Models Used

- Linear Regression – Used to predict final student scores.
- Random Forest Regressor – Used for accurate score prediction.
- Logistic Regression – Used for pass/fail classification.
- Random Forest Classifier – Used for classification tasks.
- Support Vector Machine (SVM) – Used for classification with better decision boundaries.

4. Evaluation Metrics Used

1. Accuracy

- Measures the overall correctness of the model.
- It is the ratio of correctly predicted observations to total observations.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

2. Precision

- Measures how many predicted positive cases are actually positive.
- Important when false positives are costly.

Formula:

$$Precision = \frac{TP}{TP + FP}$$

3. Recall (Sensitivity)

- Measures how many actual positive cases are correctly identified.
- Important when missing positive cases is critical.

Formula:

$$Recall = \frac{TP}{TP + FN}$$

4. F1-Score

- Harmonic mean of Precision and Recall.
- Useful when there is class imbalance.

Formula:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

5. Confusion Matrix

- A table used to describe the performance of a classification model.
- Predicted Positive Predicted Negative

Actual Positive TP FN

Actual Negative FP TN

- Helps visualize model performance in detail.

6. Classification Report

- A summary of:
 - Precision
 - Recall
 - F1-score
 - Support (number of actual samples)
 - Generated using sklearn:
- `from sklearn.metrics import classification_report`

7. ROC Curve (Receiver Operating Characteristic)

- A graph showing the performance of a classification model at different thresholds.
- Plots:
 - True Positive Rate (TPR) vs False Positive Rate (FPR)

Formulas:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

5. Results

The implemented machine learning models were trained and tested successfully. Random Forest and Logistic Regression models provided better prediction accuracy for classification tasks. The project demonstrates that machine learning can effectively predict student performance using academic and behavioral data.

6. Conclusion and Future Scope

This project successfully predicts student performance using machine learning algorithms. The system helps in identifying students who may need academic support. Future improvements may include deep learning techniques, real-time student monitoring, and deployment as a web-based application.

7. References

- [Python Documentation](#)
- [Scikit-learn Documentation](#)
- [Pandas Documentation](#)
- [NumPy Documentation](#)
- [GeeksforGeeks](#)